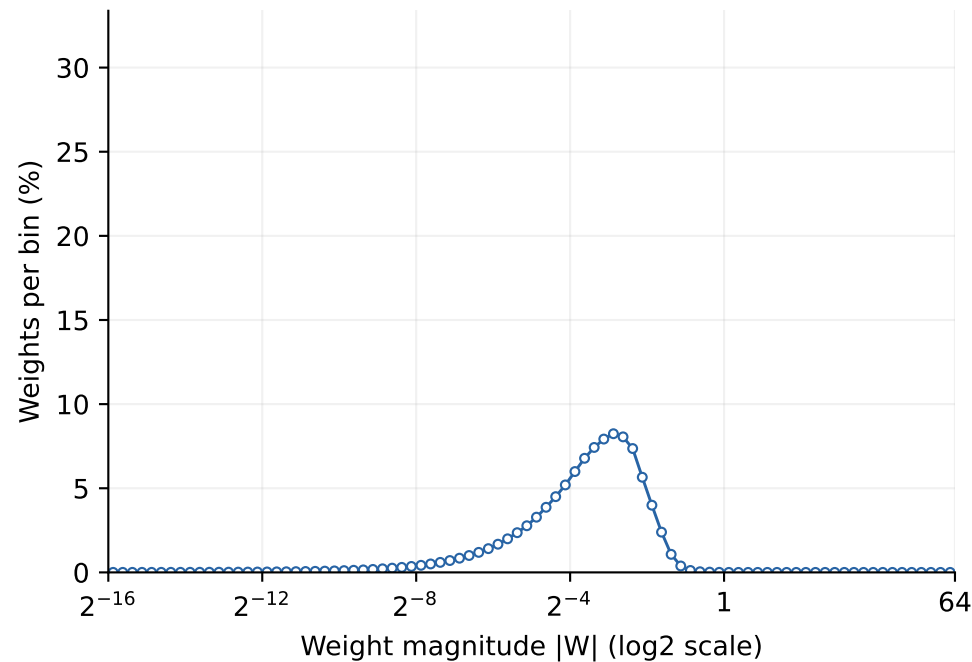


GPT-2 (124M) / Embedding / tied LM head

1 matrix tensor(s) | 38,597,376 weights | FP16 vs PoTaI HP1

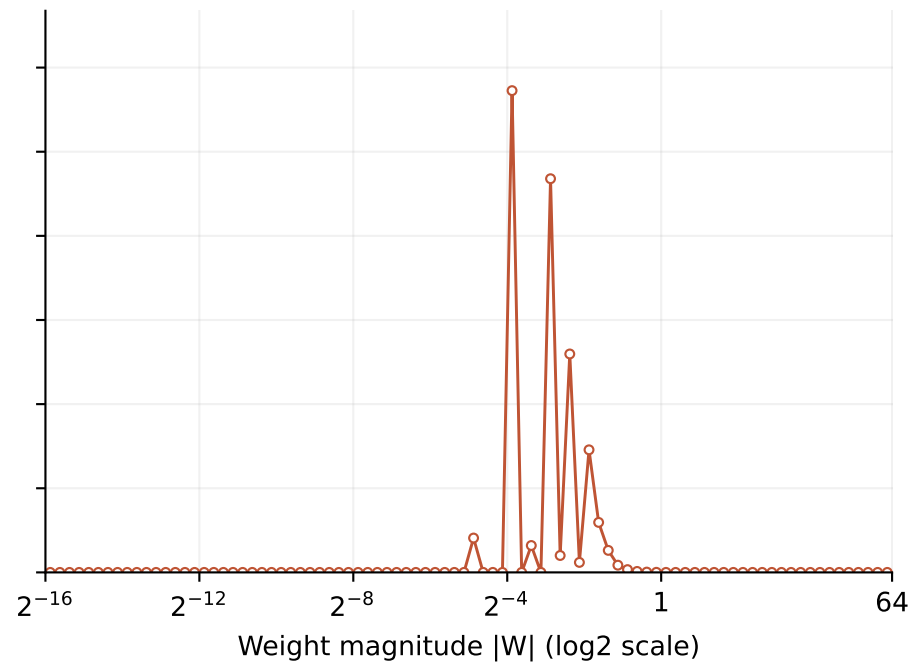
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



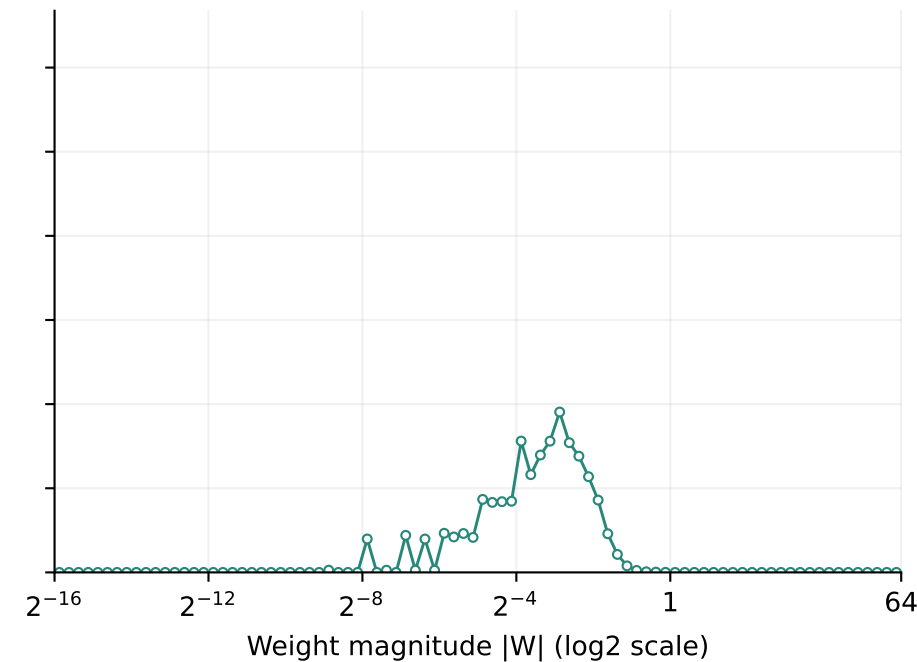
RMS 0.143696
Max |W| 1.78516
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0085%

INT4 (Q4_HP1)



RMS 0.144736
Max |W| 1.75
Zero weights 17.5101%
Relative L2 error 13.096%
SQNR 17.657 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.143752
Max |W| 1.78125
Zero weights 1.1050%
Relative L2 error 0.818%
SQNR 41.741 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: token_embd.

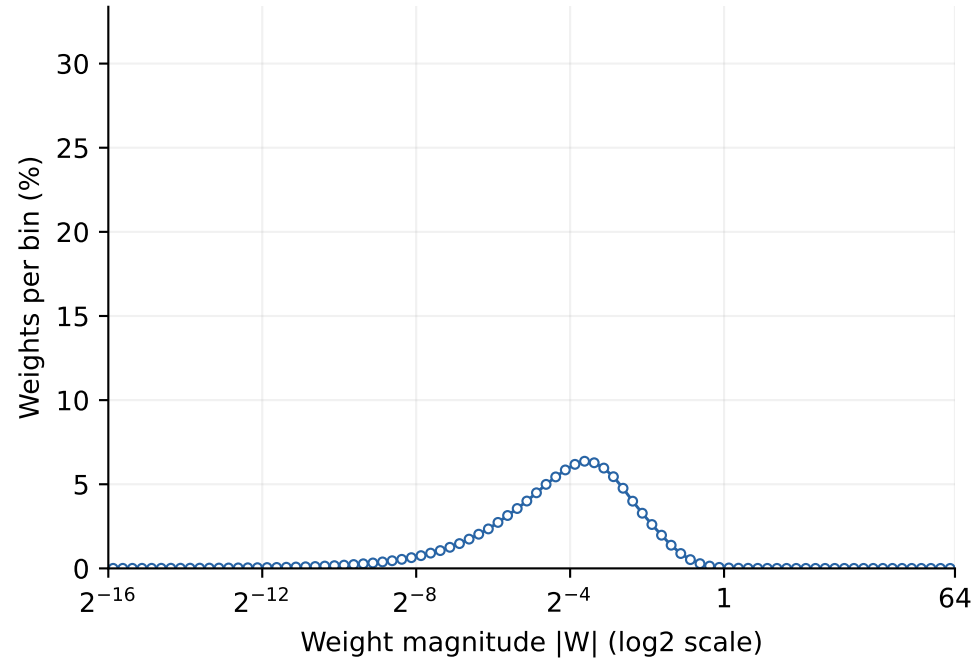
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 00

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

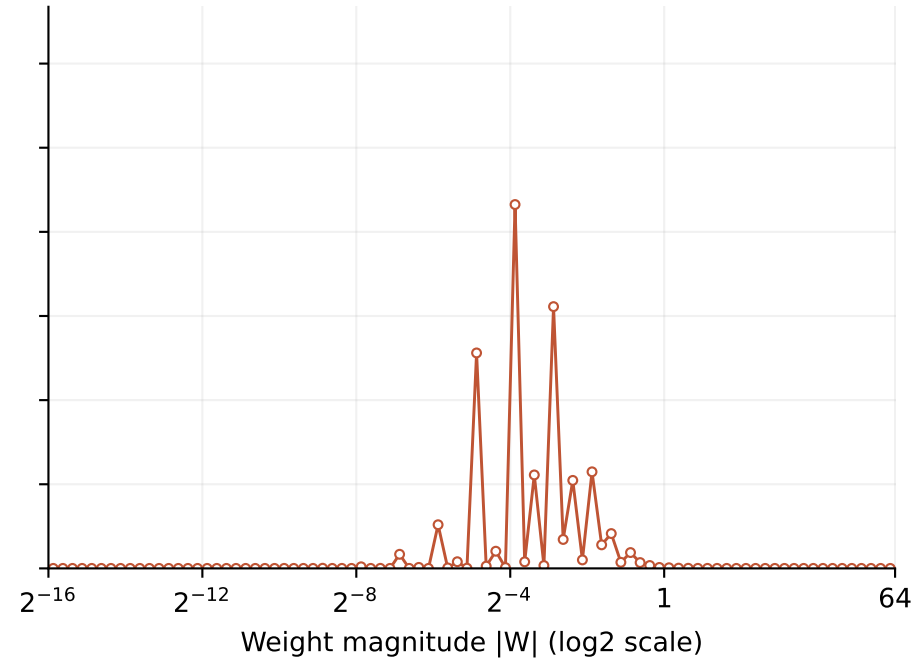
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



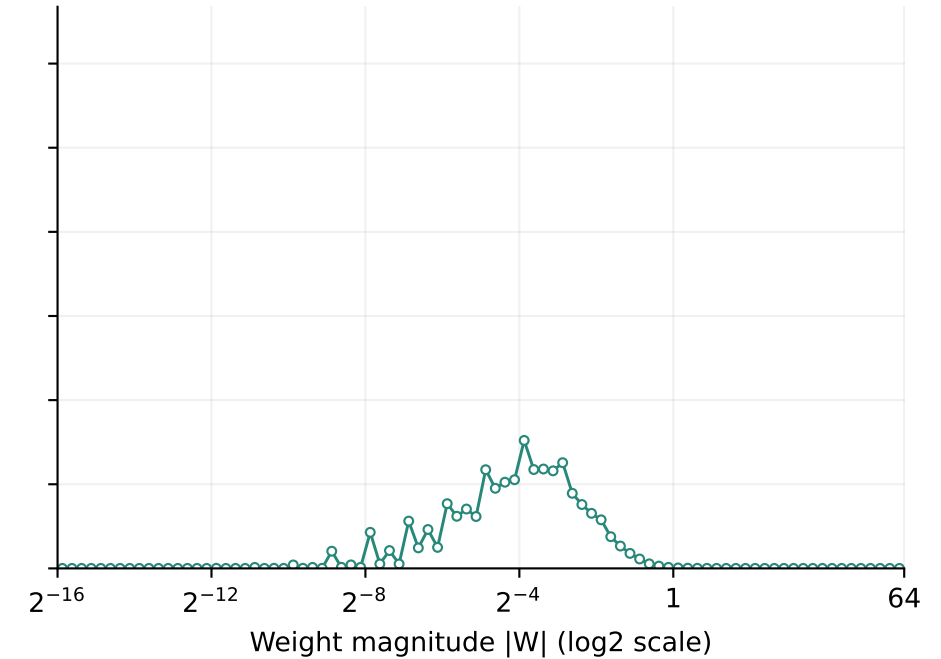
RMS 0.144902
Max |W| 6.14453
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0153%

INT4 (Q4_HP1)



RMS 0.145984
Max |W| 6
Zero weights 20.0437%
Relative L2 error 13.687%
SQNR 17.274 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.144955
Max |W| 6.125
Zero weights 1.3324%
Relative L2 error 0.873%
SQNR 41.183 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

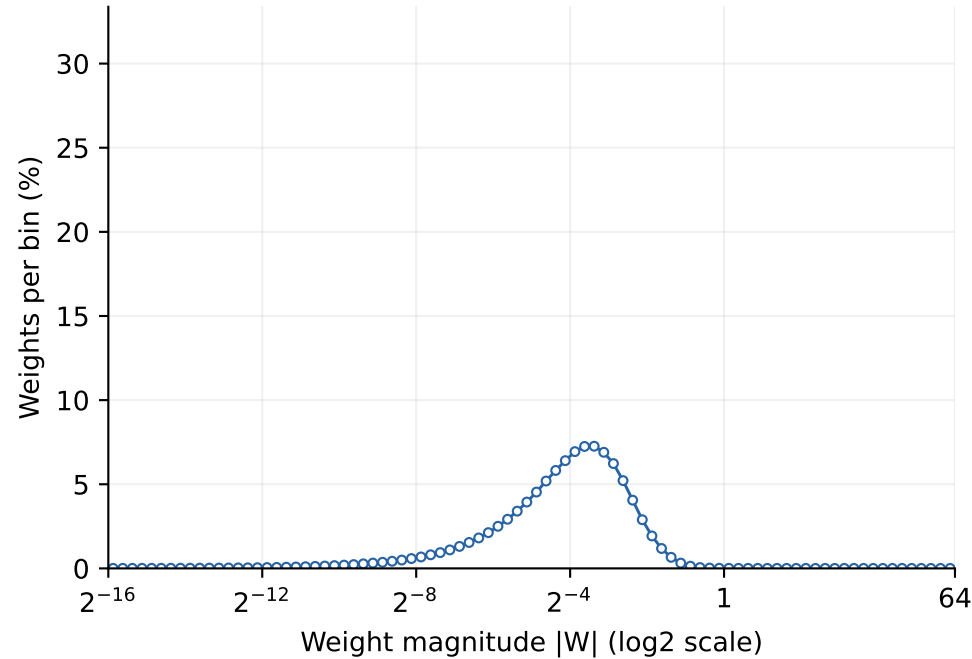
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 01

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

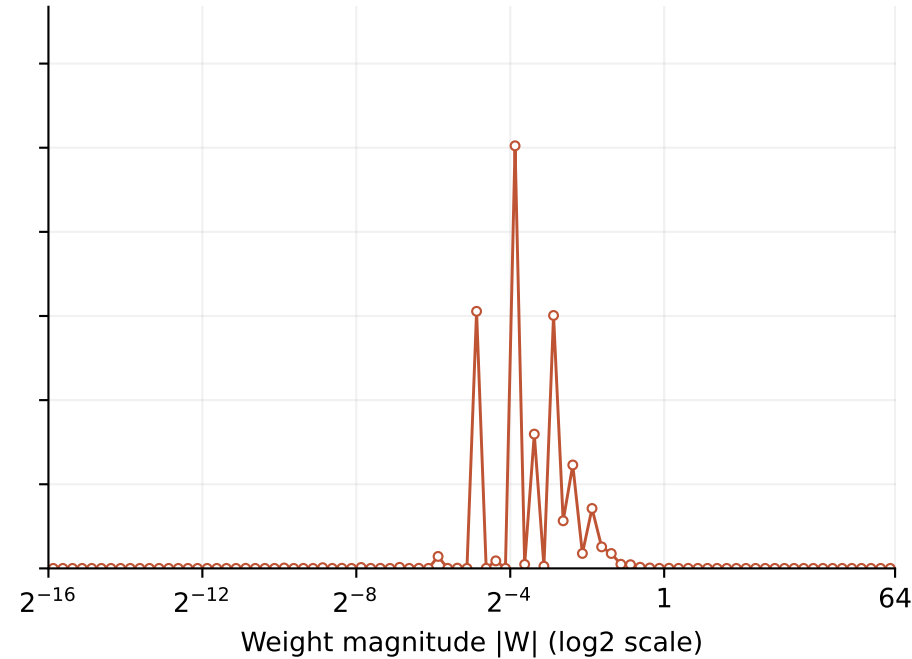
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



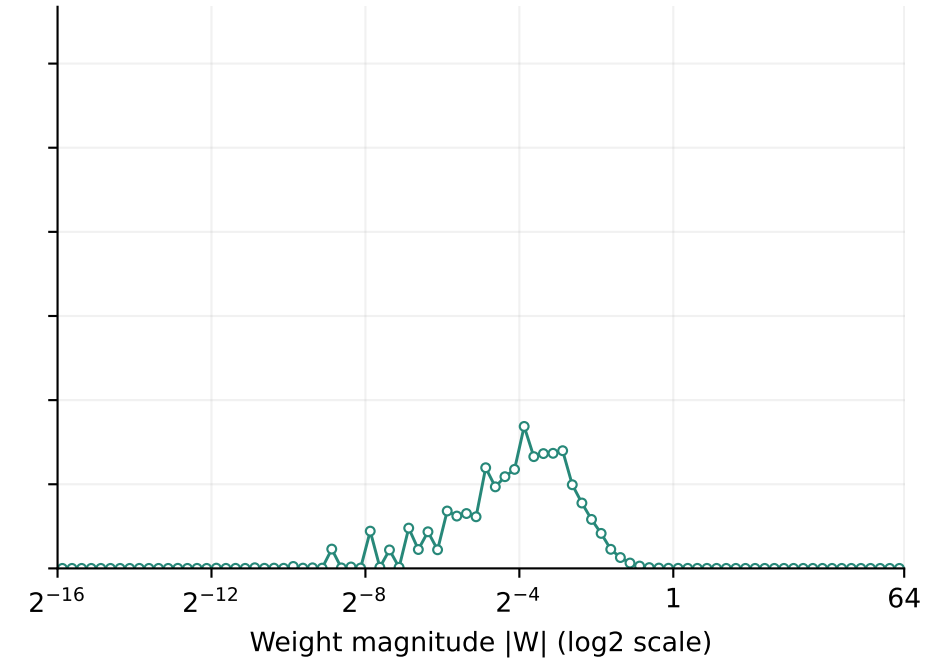
RMS 0.118321
Max |W| 13.7344
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0167%

INT4 (Q4_HP1)



RMS 0.119114
Max |W| 14
Zero weights 18.6487%
Relative L2 error 13.103%
SQNR 17.652 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.118364
Max |W| 13.75
Zero weights 1.2867%
Relative L2 error 0.833%
SQNR 41.586 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

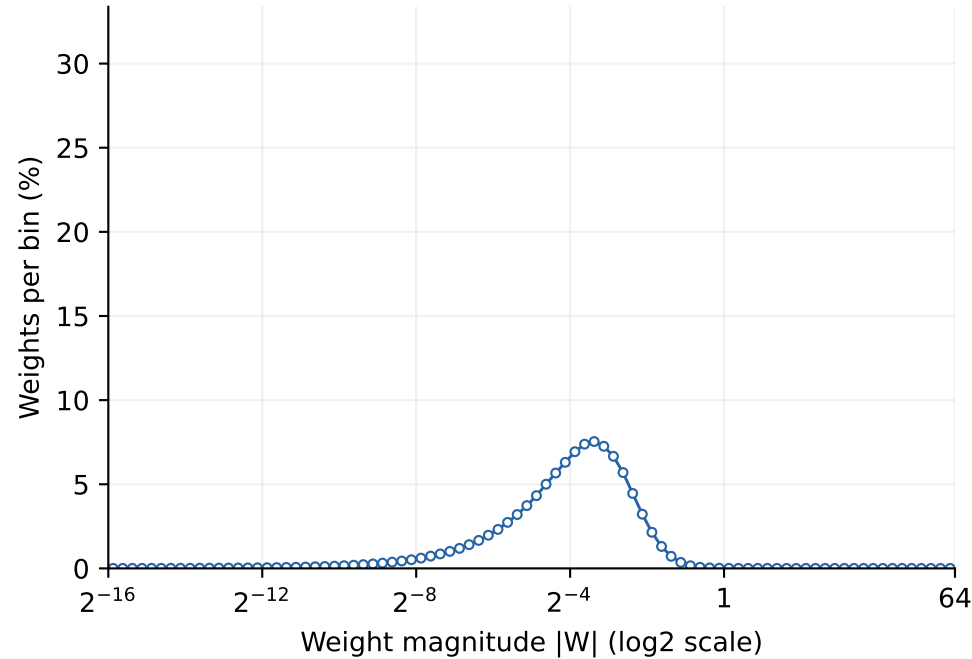
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 02

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

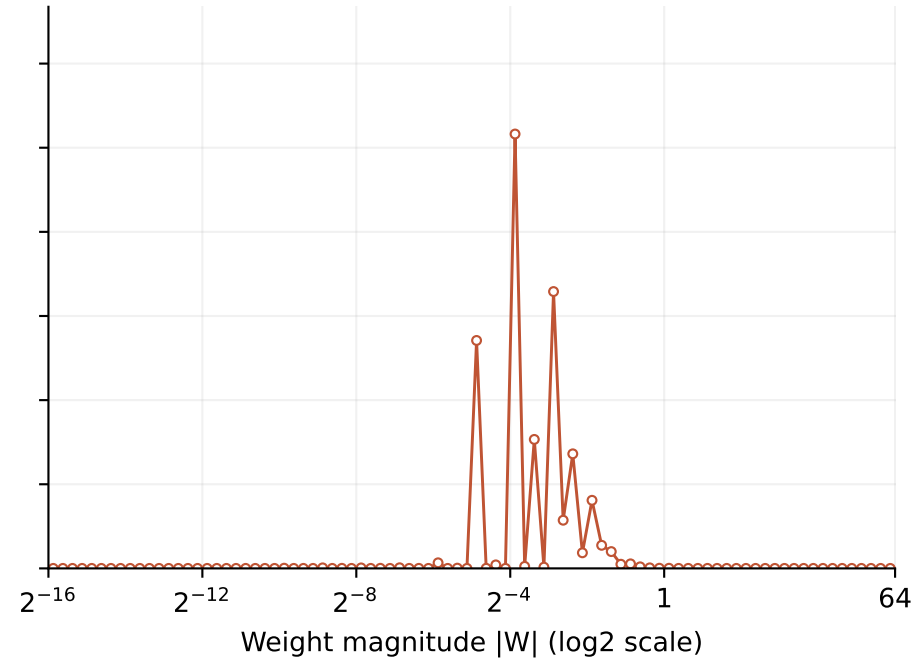
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



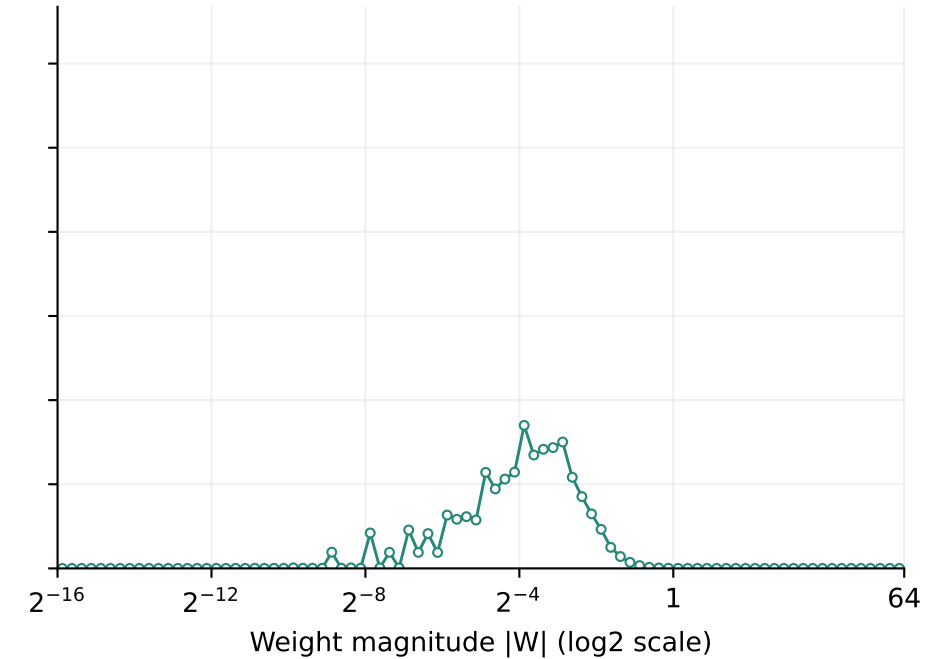
RMS 0.123346
Max |W| 15.0703
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0140%

INT4 (Q4_HP1)



RMS 0.124209
Max |W| 14
Zero weights 17.9507%
Relative L2 error 13.300%
SQNR 17.523 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.123392
Max |W| 15.125
Zero weights 1.1677%
Relative L2 error 0.838%
SQNR 41.540 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

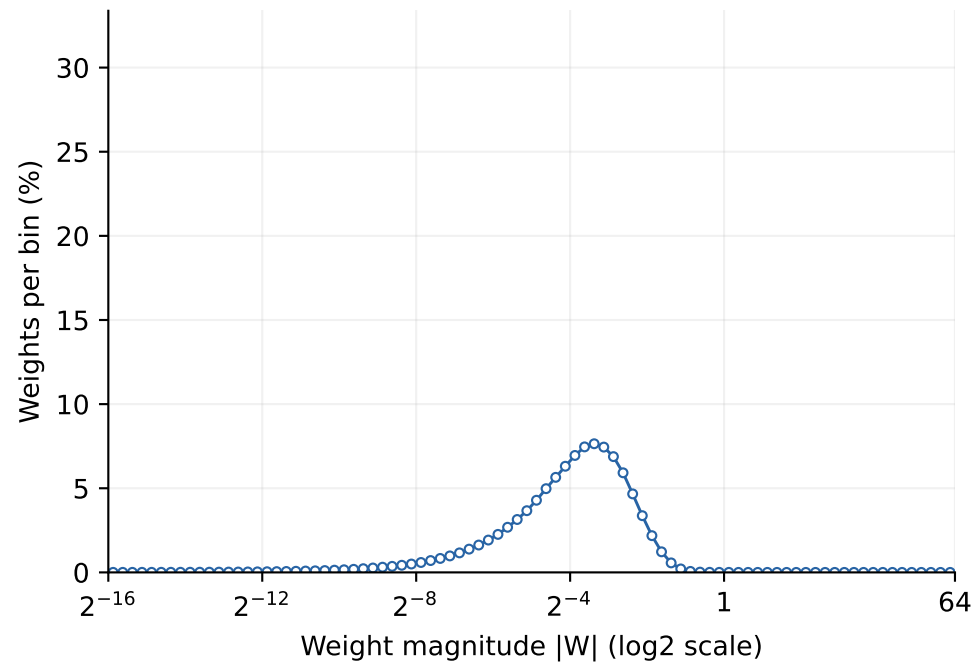
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 03

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

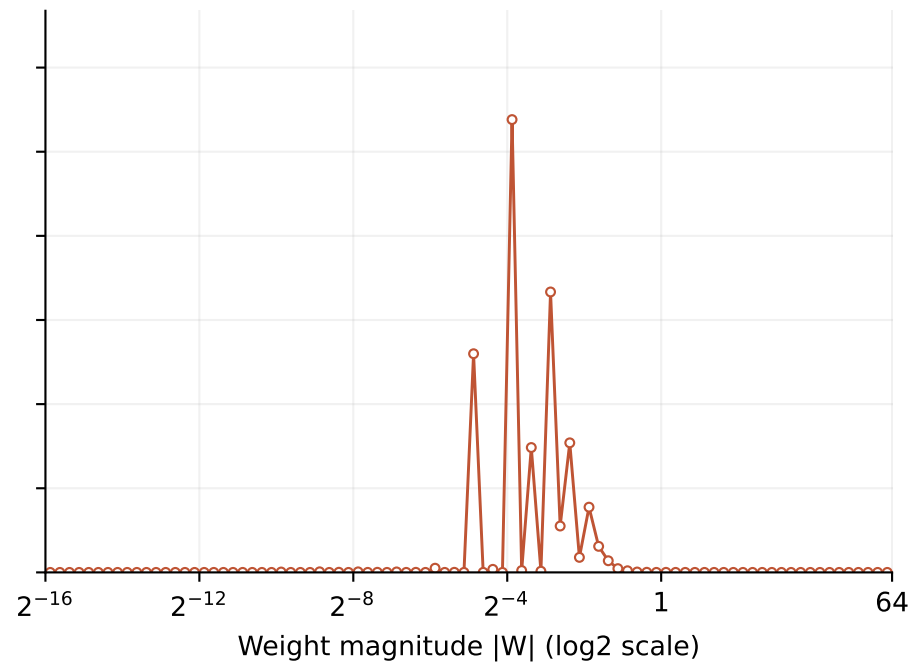
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



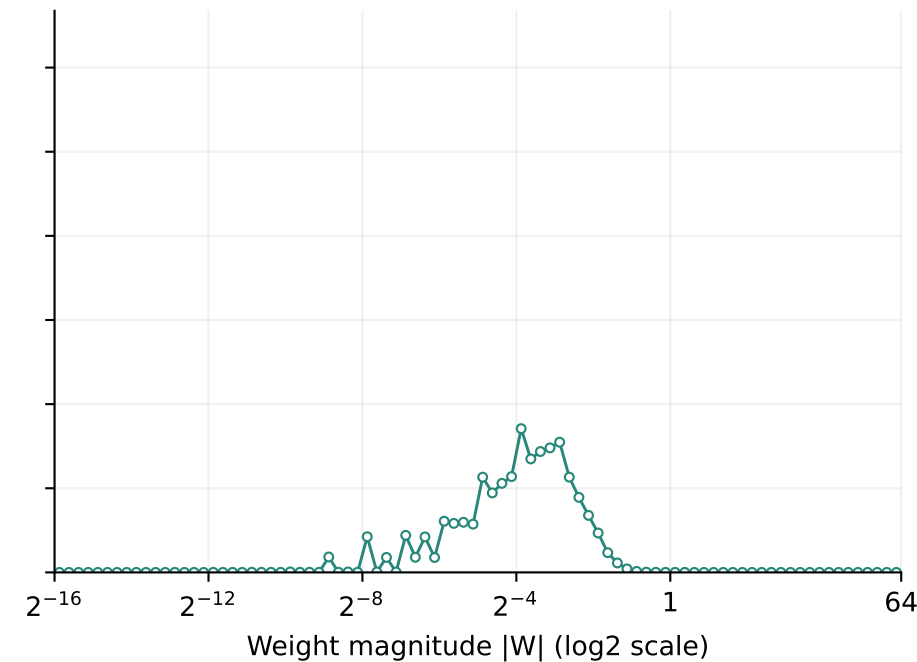
RMS 0.118453
Max |W| 17.1094
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0129%

INT4 (Q4_HP1)



RMS 0.119272
Max |W| 16
Zero weights 17.4099%
Relative L2 error 13.046%
SQNR 17.690 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.118494
Max |W| 17
Zero weights 1.1155%
Relative L2 error 0.825%
SQNR 41.673 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

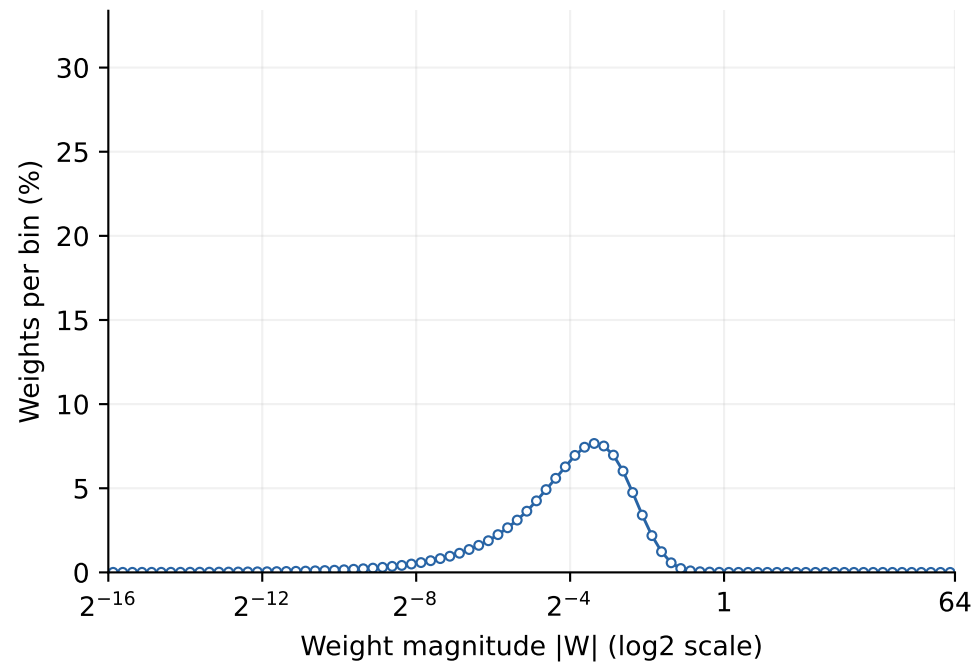
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 04

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

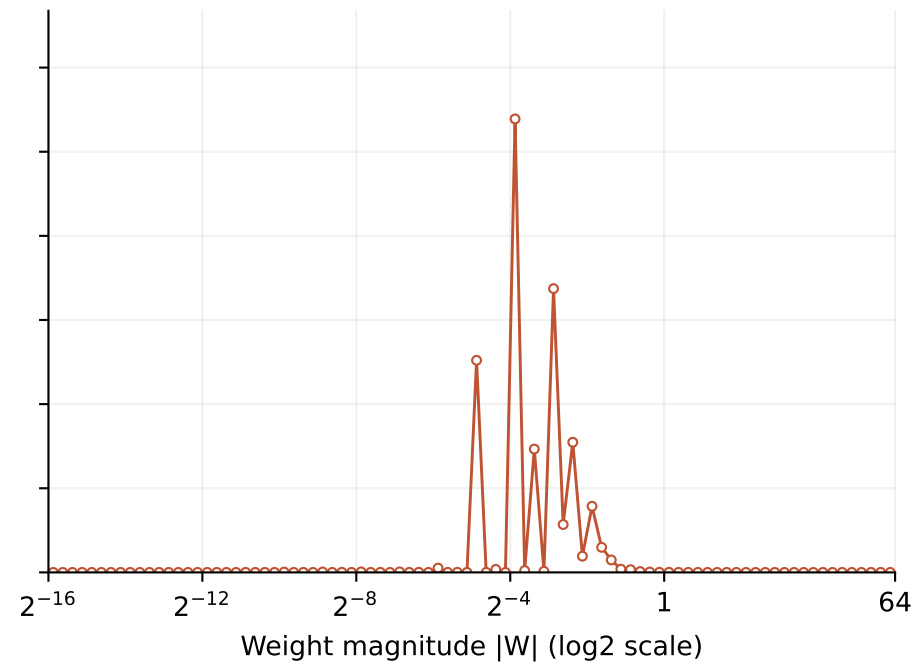
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



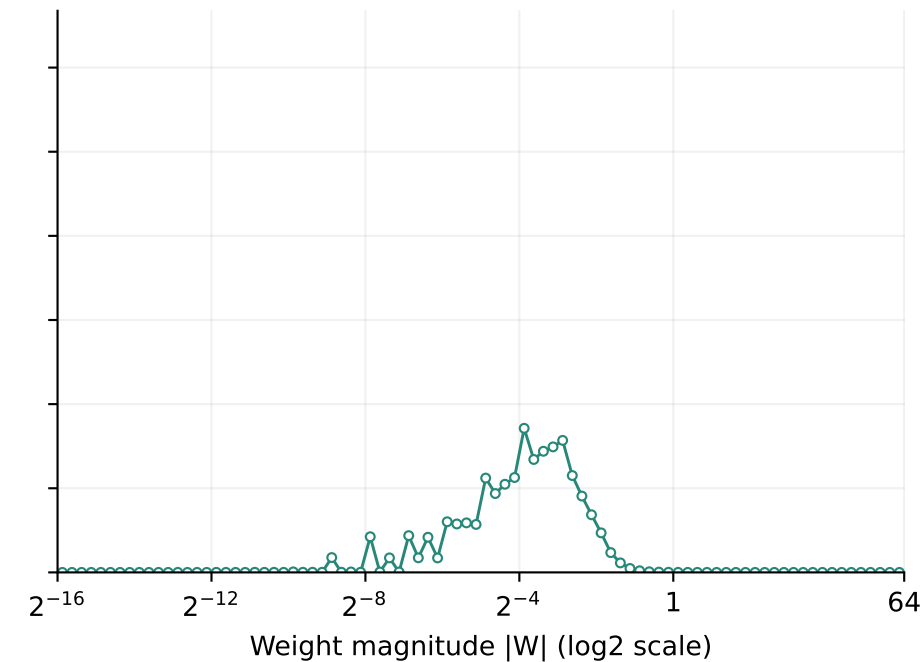
RMS 0.120224
Max |W| 4.77734
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0133%

INT4 (Q4_HP1)



RMS 0.121073
Max |W| 5
Zero weights 17.3469%
Relative L2 error 13.031%
SQNR 17.700 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.120267
Max |W| 4.75
Zero weights 1.1196%
Relative L2 error 0.815%
SQNR 41.773 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

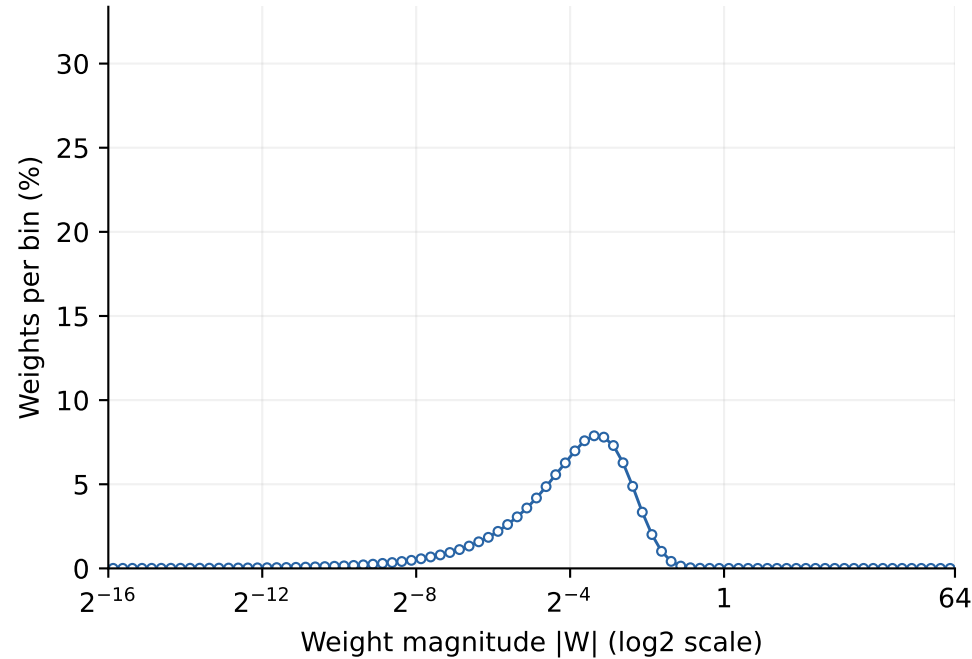
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 05

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

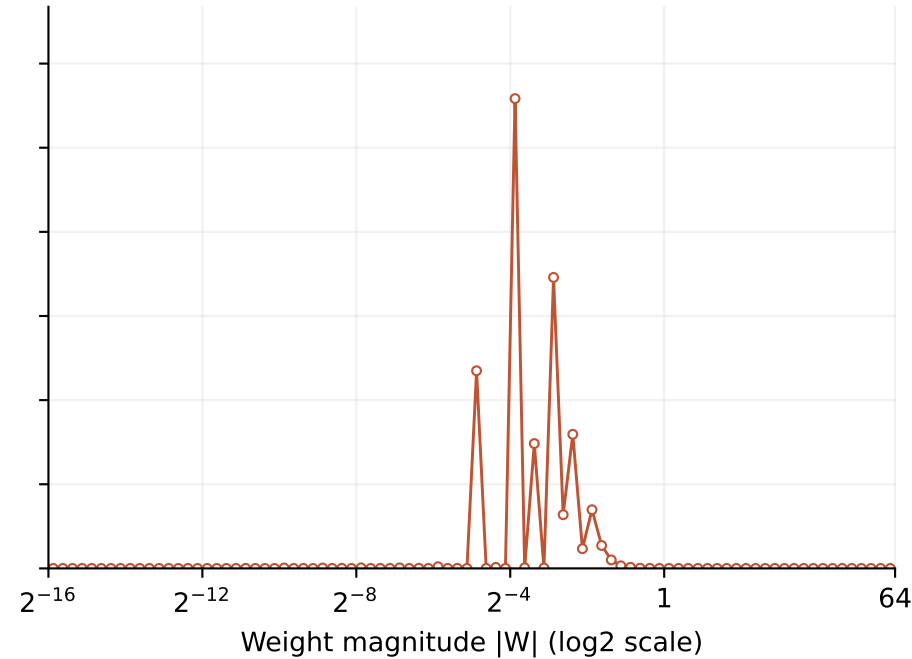
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



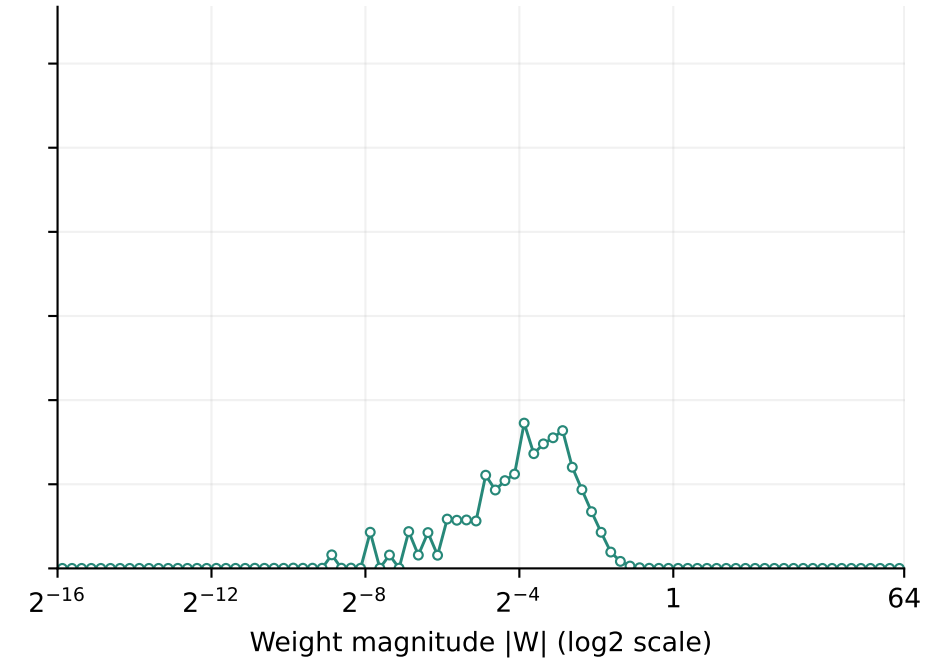
RMS 0.115527
Max |W| 2.83984
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0125%

INT4 (Q4_HP1)



RMS 0.116342
Max |W| 3
Zero weights 17.3159%
Relative L2 error 13.027%
SQNR 17.703 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.11557
Max |W| 2.84375
Zero weights 1.1076%
Relative L2 error 0.814%
SQNR 41.791 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

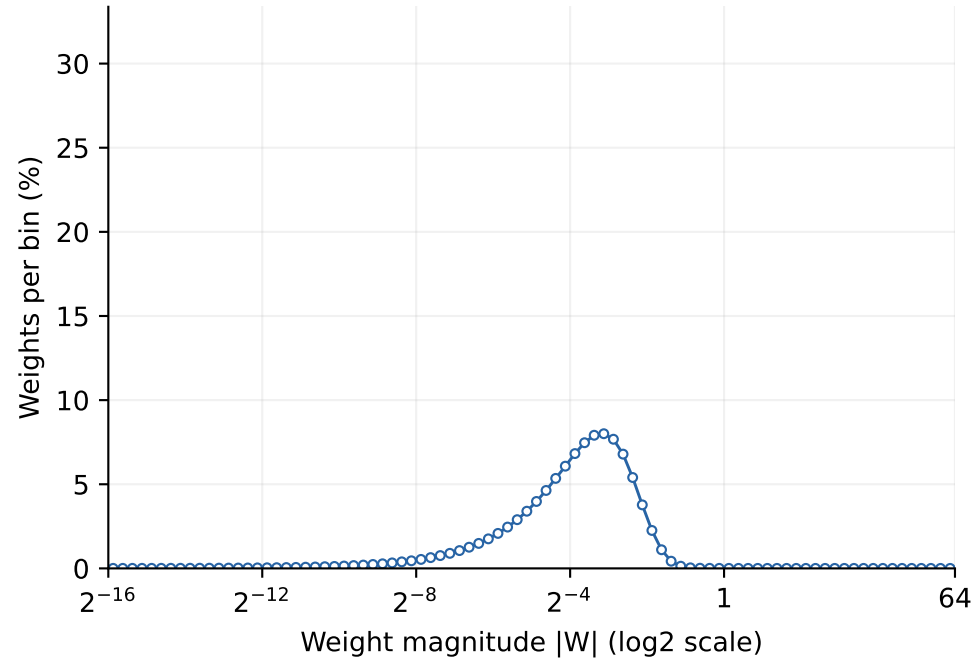
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 06

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

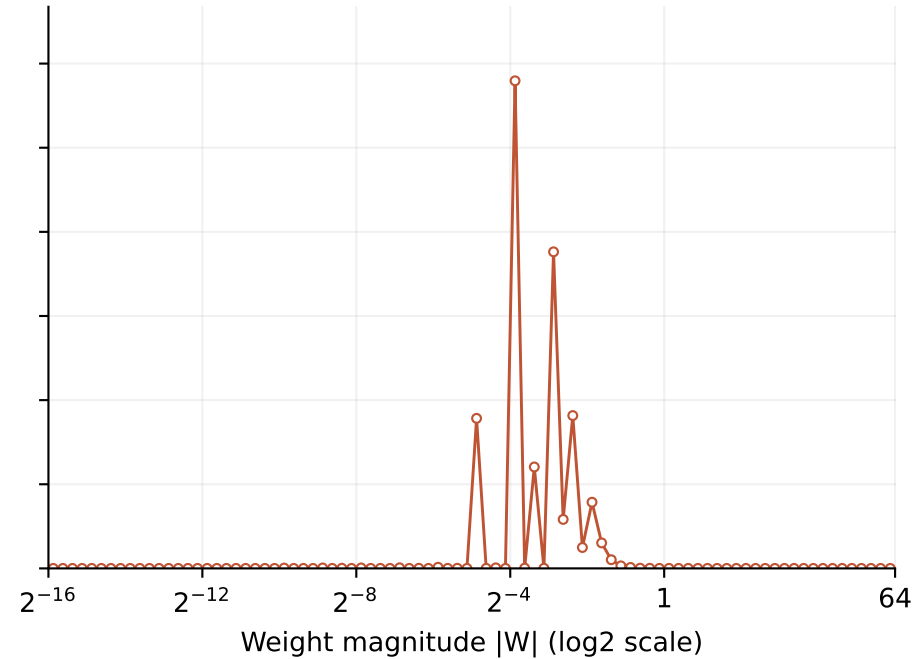
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



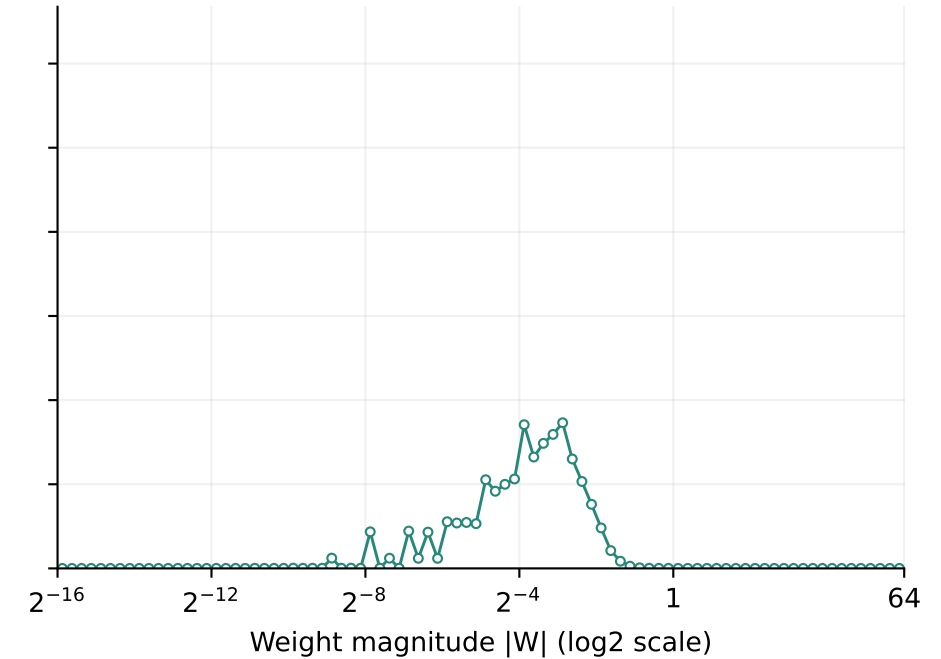
RMS 0.119437
Max |W| 2.76953
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0117%

INT4 (Q4_HP1)



RMS 0.120302
Max |W| 3
Zero weights 17.5584%
Relative L2 error 13.194%
SQNR 17.593 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.119482
Max |W| 2.78125
Zero weights 1.1198%
Relative L2 error 0.824%
SQNR 41.679 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

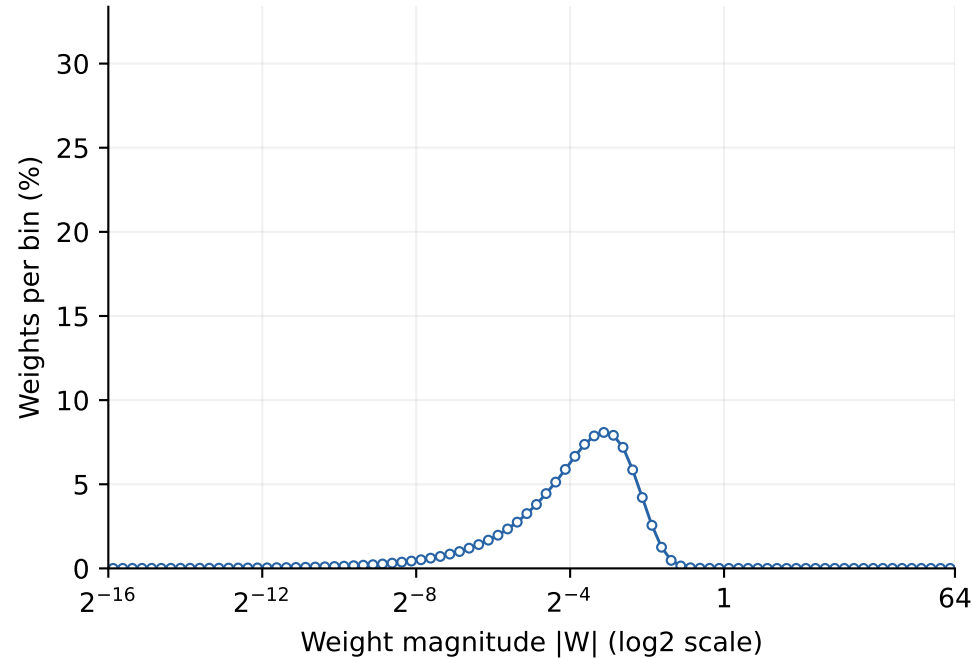
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 07

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

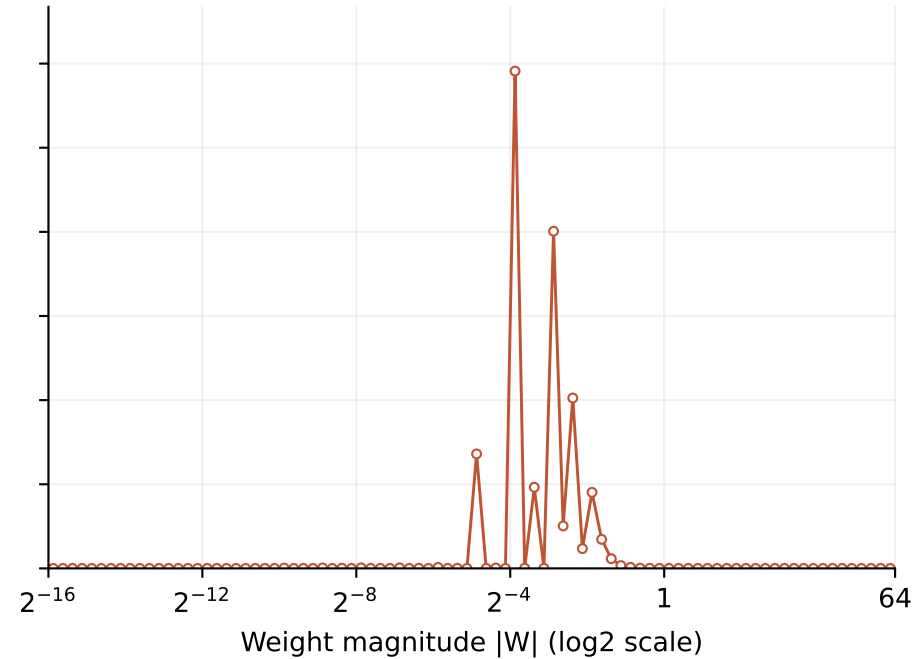
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



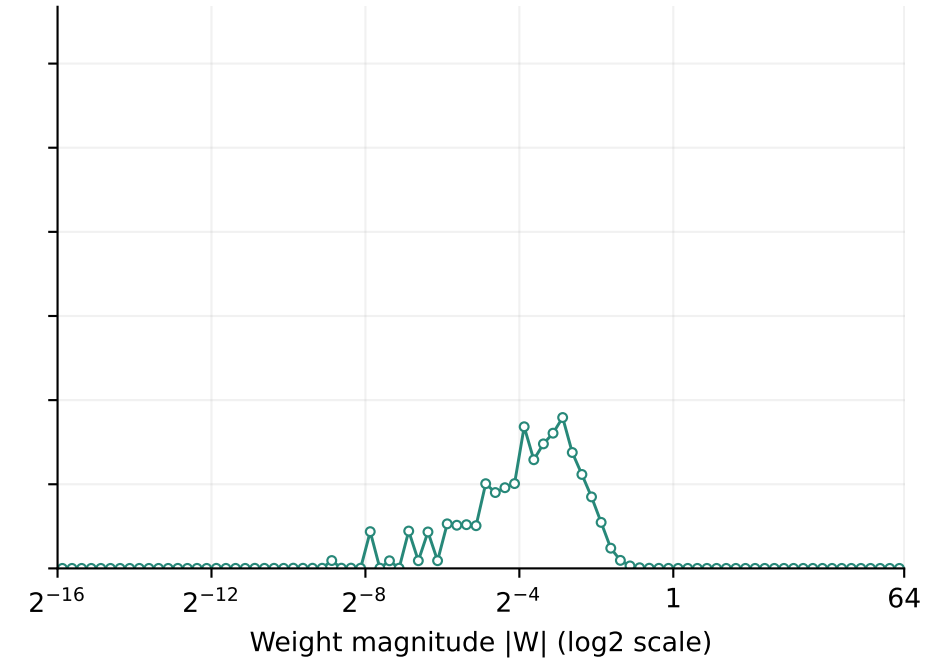
RMS 0.123579
Max |W| 4.53125
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0111%

INT4 (Q4_HP1)



RMS 0.124513
Max |W| 5
Zero weights 17.6186%
Relative L2 error 13.249%
SQNR 17.556 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.123626
Max |W| 4.5625
Zero weights 1.1217%
Relative L2 error 0.828%
SQNR 41.643 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

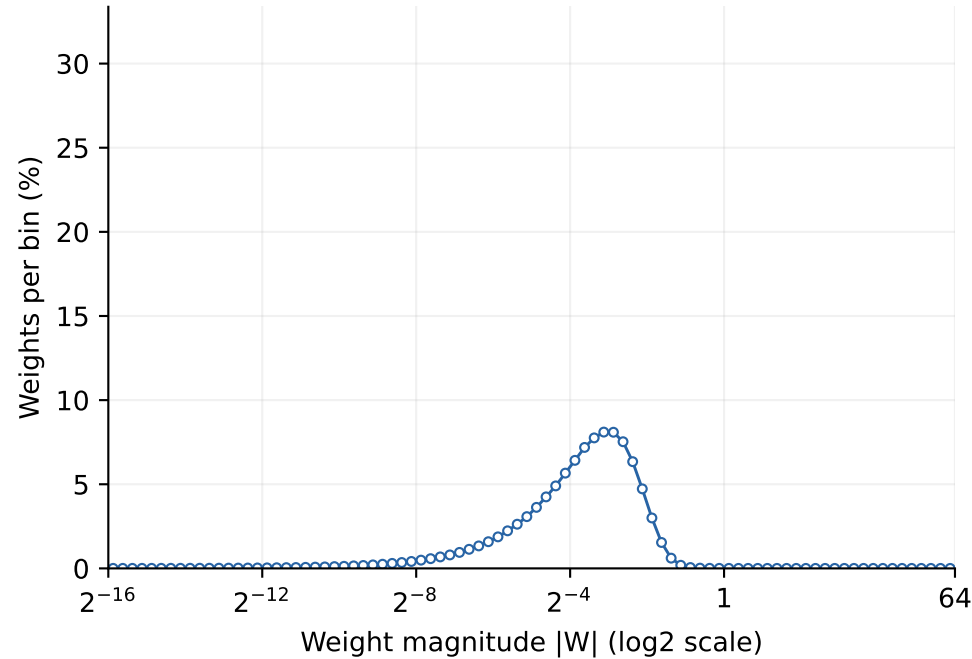
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 08

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

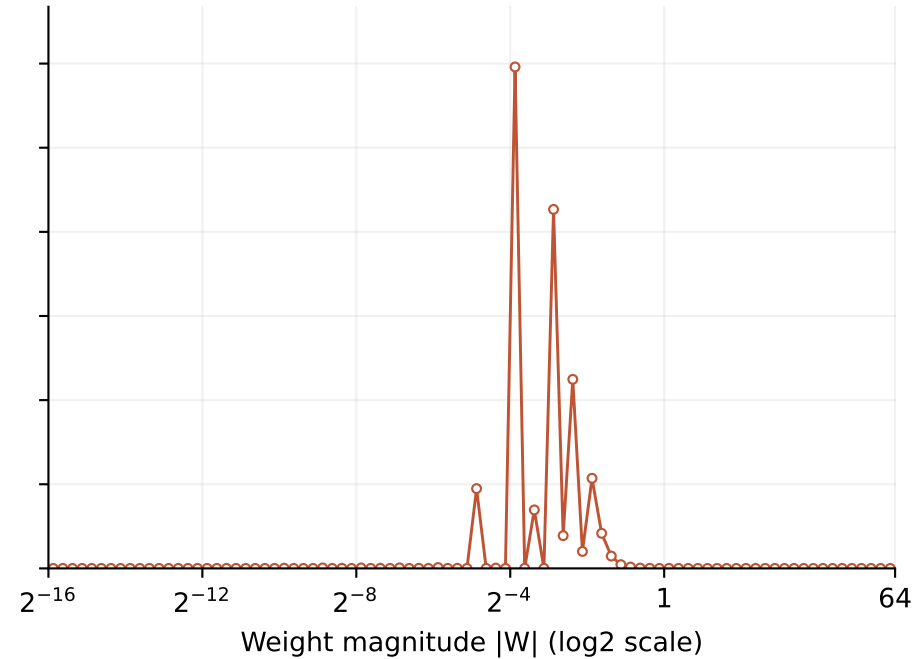
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



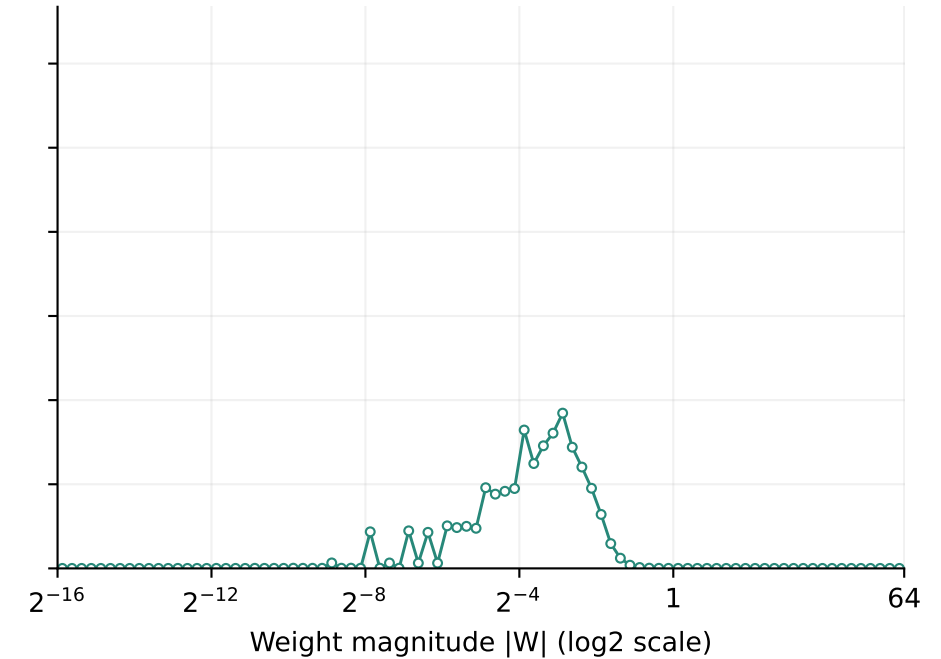
RMS 0.129571
Max |W| 5.375
Zero weights 0.0001%
FP16 reference
Nonzero mass outside x-axis 0.0109%

INT4 (Q4_HP1)



RMS 0.130566
Max |W| 5
Zero weights 17.7225%
Relative L2 error 13.283%
SQNR 17.534 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.129619
Max |W| 5.375
Zero weights 1.1240%
Relative L2 error 0.830%
SQNR 41.620 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

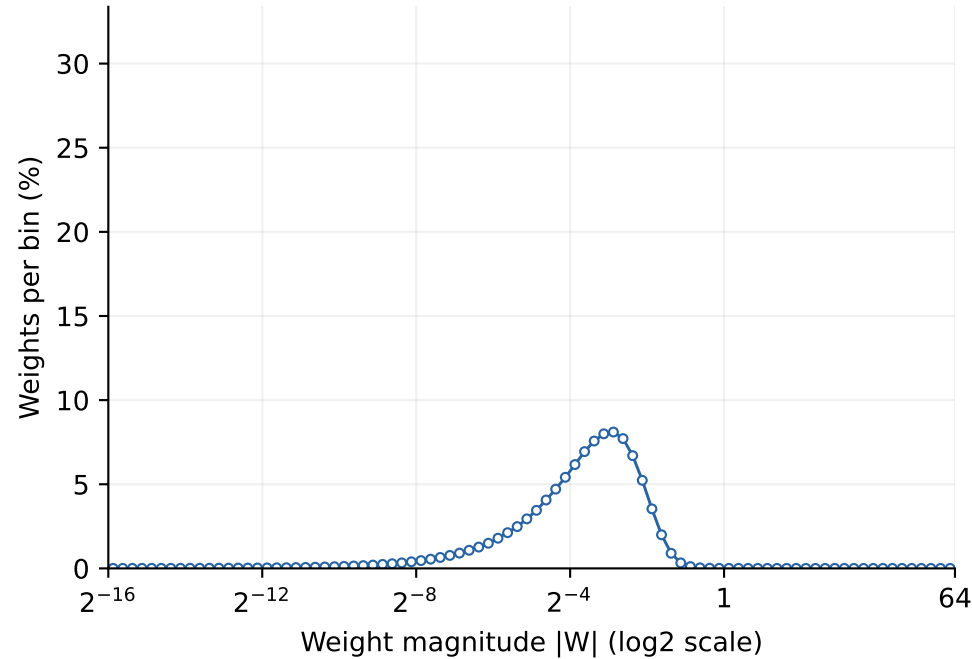
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 09

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

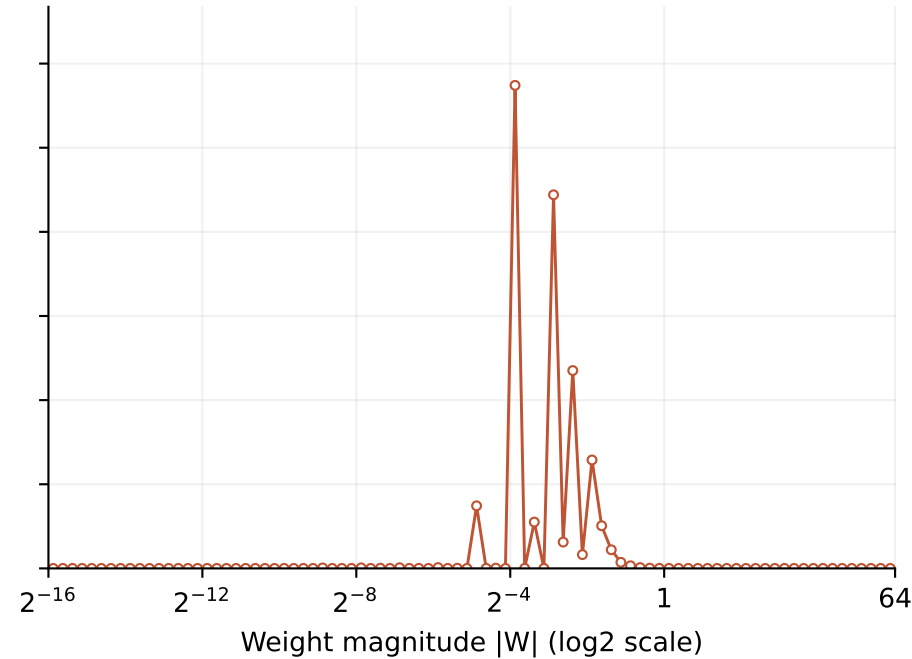
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



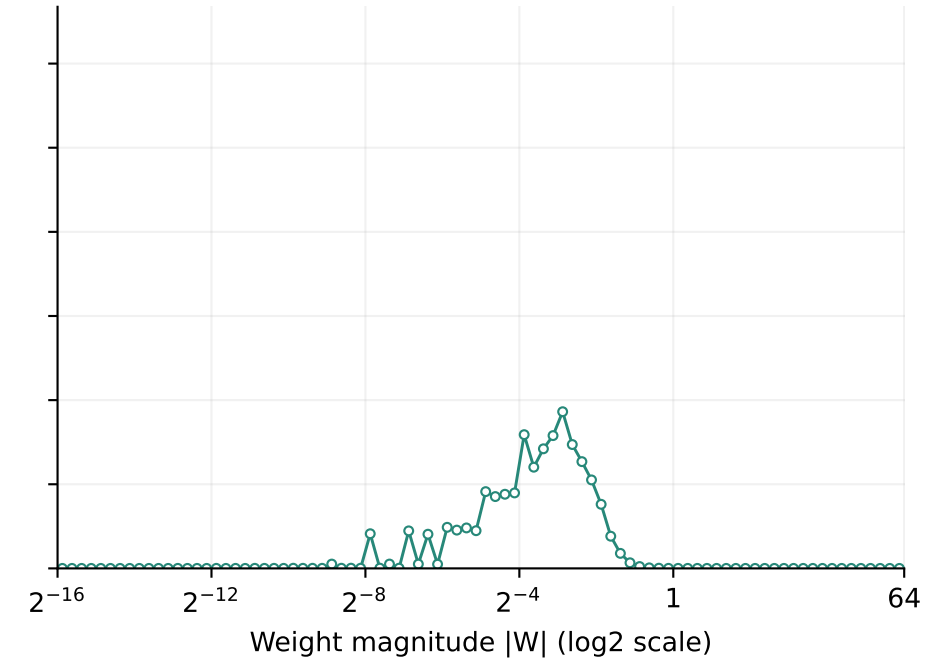
RMS 0.138105
Max |W| 5.48828
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0098%

INT4 (Q4_HP1)



RMS 0.139131
Max |W| 5
Zero weights 17.5851%
Relative L2 error 13.183%
SQNR 17.600 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.138155
Max |W| 5.5
Zero weights 1.1226%
Relative L2 error 0.823%
SQNR 41.689 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

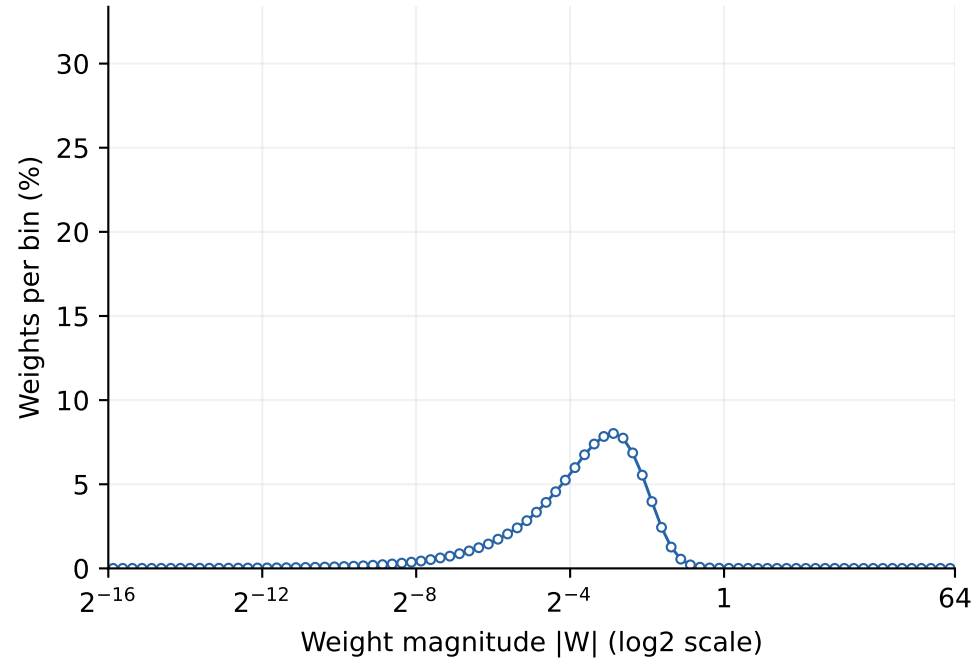
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 10

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

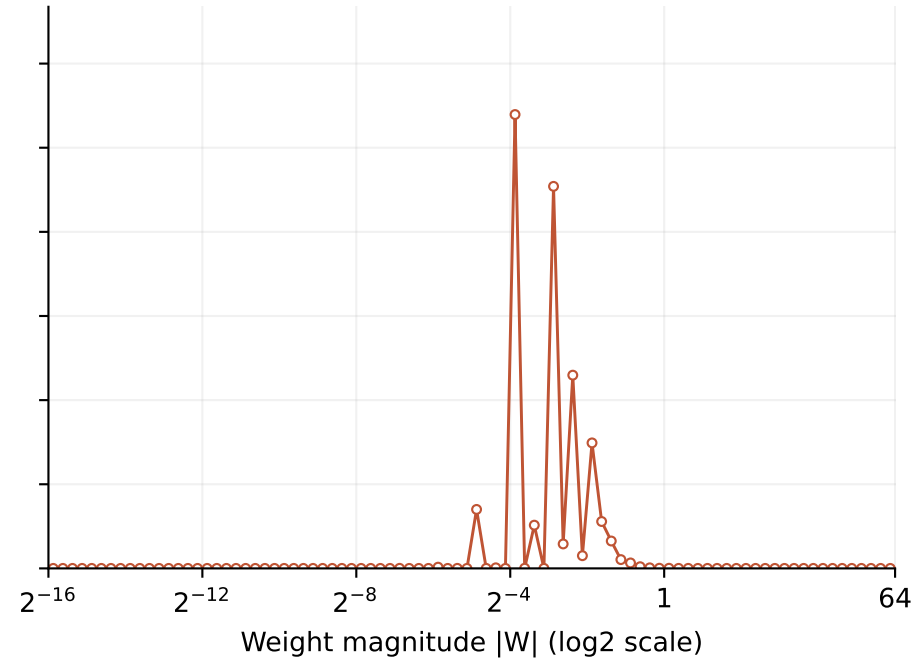
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



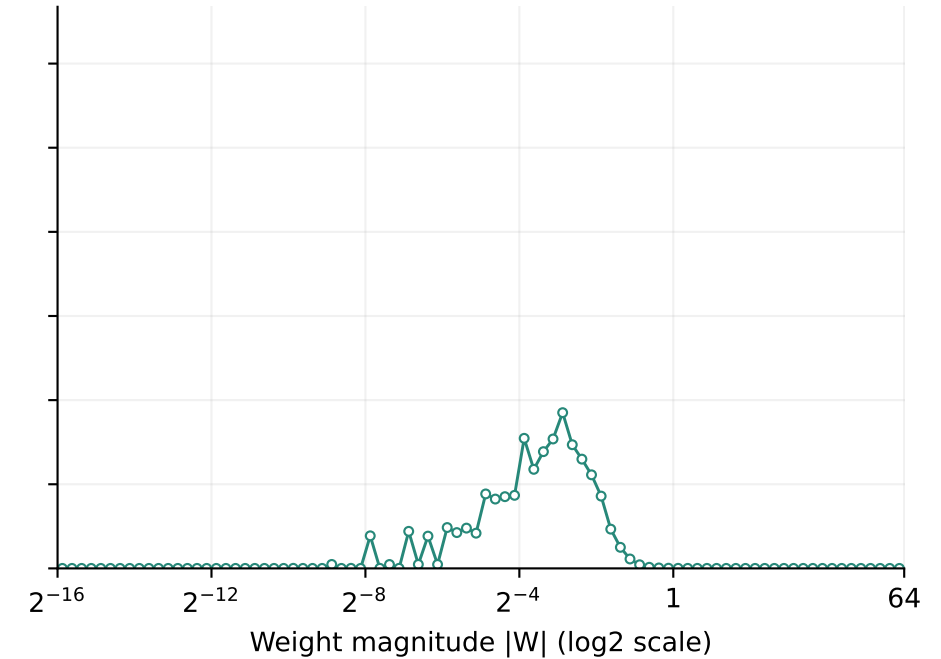
RMS 0.147704
Max |W| 11.0469
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0088%

INT4 (Q4_HP1)



RMS 0.148785
Max |W| 12
Zero weights 17.5505%
Relative L2 error 13.261%
SQNR 17.549 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.147758
Max |W| 11
Zero weights 1.1147%
Relative L2 error 0.831%
SQNR 41.612 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

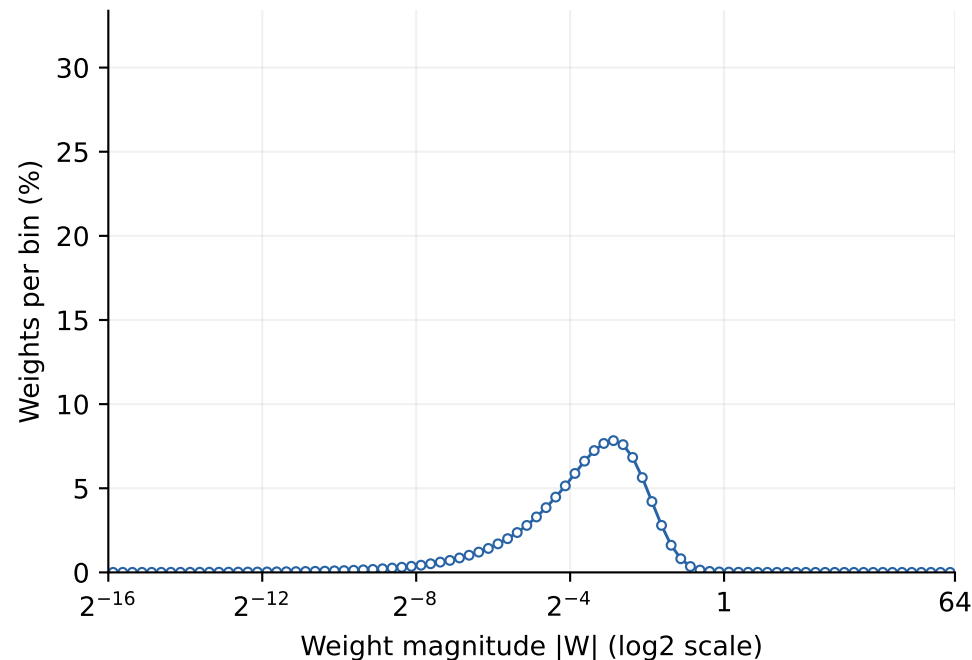
Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.

GPT-2 (124M) / Transformer block 11

4 matrix tensor(s) | 7,077,888 weights | FP16 vs PoTal HP1

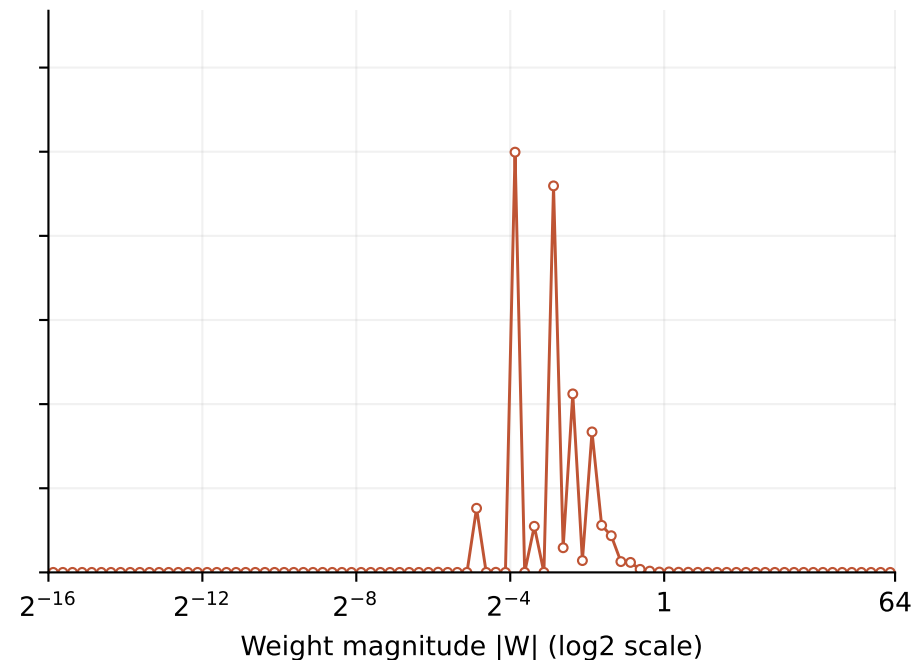
Observed histogram points joined by straight lines. Identical axes across all layers and precisions of this model.

FP16



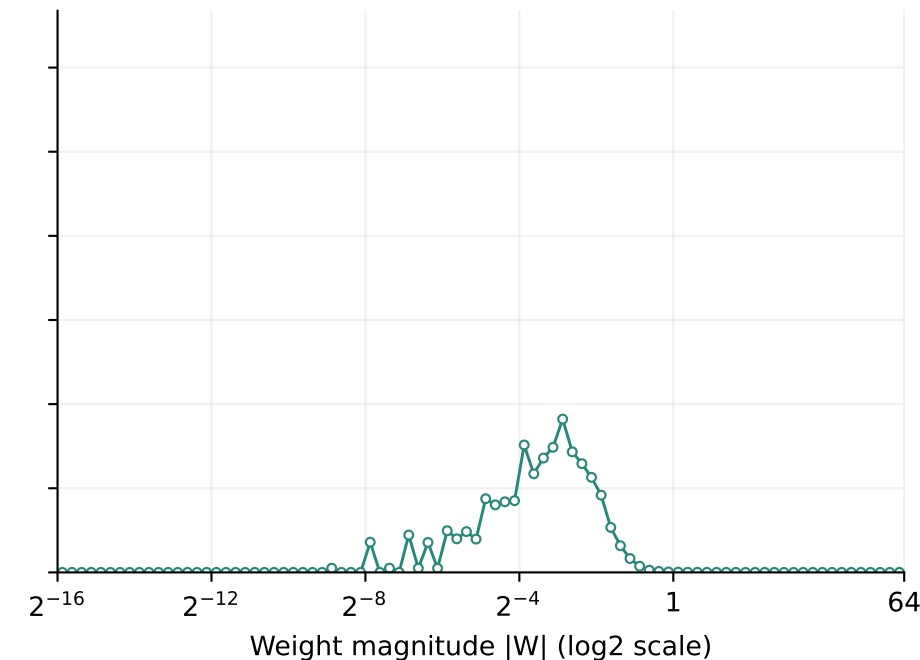
RMS 0.160051
Max |W| 9.21094
Zero weights 0.0000%
FP16 reference
Nonzero mass outside x-axis 0.0091%

INT4 (Q4_HP1)



RMS 0.161153
Max |W| 10
Zero weights 17.8134%
Relative L2 error 13.511%
SQNR 17.386 dB
Nonzero mass outside x-axis 0.0000%

INT8 (Q8_HP1)



RMS 0.160109
Max |W| 9.25
Zero weights 1.1325%
Relative L2 error 0.859%
SQNR 41.322 dB
Nonzero mass outside x-axis 0.0000%

Included matrices: attn_qkv, attn_output, ffn_up, ffn_down.

Each point: 100 x bin count / total weights. Bin width: 0.25 log2 units. Zero is reported separately. Norm/bias excluded.